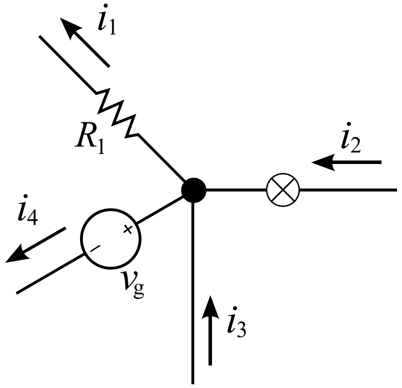
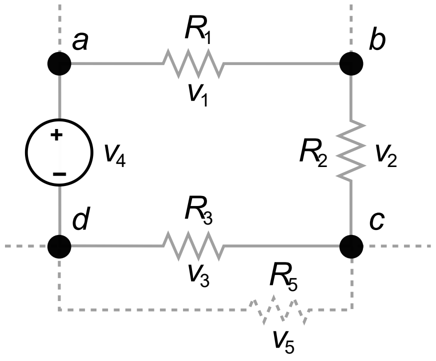
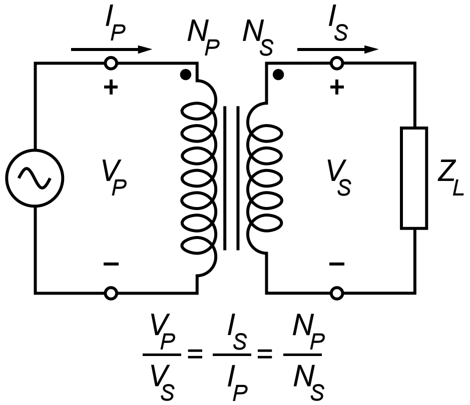
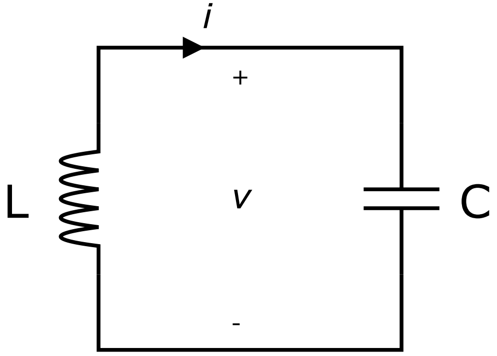
Hochschule Kaiserslautern
University of Applied Sciences

Introduction to Electrical Engineering









Overview

1. Physical Principles and Basic Concepts
2. Direct Current Circuits
3. Electric and Magnetic Fields
4. Alternating Current Technology
5. **Wrap-Up and Outlook**

Wrap-Up and Outlook

- Summary using Applications

- Outlook
  - Semiconductor Devices and Electronics
  - Energy Systems
  - Information Systems
  - Control Systems
  - Embedded Systems

Fuse and Circuit Breaker

protect circuits from **excess current**, which is usually caused by:

- overload (more devices than the circuit can handle)
- short circuit (very low resistance path)

typically 16 A for power outlets:

- maximum continuous power (assuming 230 V): P = V I = 230 V 16 A = 3680 W
- this is enough to power several devices at once, like: 1–2 kettles, 1 microwave, some small appliances together
- Beyond 16 A, the circuit breaker trips to prevent overheating and fire risk.

Residual Current Device (RCD)

circuit for fault current measurement (30 mA)

to ensure electrical safety

Capacitive Touchscreen

Inductive Sensor

Semiconductor Devices and Electronics

- diodes and transistors — switching and amplification
- operational amplifiers — building blocks for analog circuits
- analog circuit design — filters, amplifiers, signal conditioning
- digital electronics — logic gates

Energy Systems

generation, conversion, and distribution of electrical power

- transformers and power converters
- power electronics

Information Systems

transmitting and processing information

signal processing
- time-domain vs. frequency-domain analysis
- Fourier and Laplace transforms

communication systems
- amplitude/frequency modulation (AM/FM)
- digital modulation
- noise, bandwidth, and filtering concepts
- communication channels and data transmission

Control Systems

- feedback, stability, and dynamic response
- root locus, Bode, and Nyquist techniques
- PID control and real-world applications (motors, robotics)

Embedded Systems

hardware–software integration

- microcontrollers and digital systems
- sensors, actuators, and real-time control